

David Elijah Umaru

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SUMMARY

Computer Engineering senior (B.S., May 2026, GPA 3.94) with a Minor in Computer Science, applying to graduate study in Computer Engineering. Experience building real-time control software, automation pipelines, and internal tooling in Python, C++, and Linux environments — from a 1kHz motor-control loop to an MQTT-based dashboard serving live sensor data. Comfortable owning a project from architecture through documentation and deployment.

EDUCATION

Colorado State University — Fort Collins, CO

May 2026

B.S. Computer Engineering, Minor in Computer Science — Concentration: Embedded Systems

GPA: 3.94/4.0 (Dean's List, 6 semesters)

EXPERIENCE

HPC Developer Intern

June 2025 – Present

Data Science Research Institute, Colorado State University

- Supported and troubleshooted high-availability Linux/HPC systems, improving computational reliability and reducing researcher downtime.
- Assisted with system validation and commissioning during a new HPC data-center rebuild, including configuration checks and operational testing.
- Authored internal technical documentation (GitHub, Read the Docs, Freshservice) and trained faculty/students on Linux CLI workflows and job scheduling, improving support and onboarding efficiency ~50%.

Automation Intern

August 2022 – December 2022

Dept. of Microbiology, Immunology, and Pathology, Colorado State University

- Developed Python automation scripts using the Opentrons API, increasing laboratory throughput ~50%.
- Automated dilution, titration, and fluorescence workflows with an emphasis on repeatability and error reduction.
- Collaborated with researchers to integrate magnetic bead protein purification on the Opentrons Flex platform.

IT Student Support & A/V Media Specialist

June 2023 – Present

College of Business, Colorado State University

- Resolved IT incidents for 275+ faculty/staff, deployed/imaged 150+ Windows systems, and led Crestron/AV podium renovations, training 50+ end users.

PROJECTS

RamBots — Real-Time Motor Control Software (Quadrupedal Robot)

Aug 2025 – Present

Python, C++, Teensy 4.1, ODrive, CAN bus

- Rewrote the robot's C++ control firmware, replacing blocking serial I/O with interrupt-driven UART to raise the control loop from 20Hz to 1kHz, and built an inverse-kinematics solver with a modular Leg/Axis/ODrive class architecture.
- Parsed CAN telemetry from six ODrive motor controllers; built Python configuration/telemetry tooling that let 6 new members calibrate motors independently and diagnosed a fault that raised system uptime from 60% to 98%.
- Achieved <2° joint angle error and <50ms command latency across 5+ live demonstrations with zero mid-presentation failures.

RamBots Documentation Platform & Knowledge Base

2025

Read the Docs, Sphinx, reStructuredText, GitHub CI

- Built a version-controlled documentation platform (Read the Docs + GitHub, auto-build on commit) covering hardware, software architecture, and troubleshooting for a 15+ contributor project.
- Authored full class/function API references and a 4-module onboarding curriculum, cutting new-member onboarding from 4–6 weeks to 2 weeks and reaching 200+ views in one semester.

Energy Management System for Room Heaters

Sept 2024 – Dec 2024

Python, Flask, MQTT, TensorFlow Lite, Raspberry Pi

- Built an edge-deployed energy optimization system processing 20,000+ time-series data points with an MQTT-based real-time pipeline and a Flask dashboard running at <100ms latency, deployed via TensorFlow Lite on a Raspberry Pi with no cloud dependency.

5-DOF Robotic Arm — Control Software & HMI

Aug 2025 – Present

Android, ESP32-S3, JavaScript/HTML

- Developed an Android HMI application and a JavaScript/HTML web interface for real-time, Wi-Fi/serial control of a 5-DOF robotic arm's ESP32-S3 controller, on a 6-member multidisciplinary senior design team.

TECHNICAL SKILLS

Languages: Python, C++, Java, C, MATLAB, VHDL

Frameworks & Tools: Linux, Git/GitHub, Flask, TensorFlow / TensorFlow Lite, MQTT, Node-RED, VS Code, IntelliJ IDEA

Hardware & Embedded: Raspberry Pi, ESP32, Teensy 4.1, KiCad (PCB design), CAN bus, UART, SPI

RELEVANT COURSEWORK

Data Structures & Algorithms, Embedded Systems and Machine Learning, Cybersecurity, Digital System Design, Signals and Systems, VLSI Design

LEADERSHIP & ACTIVITIES

- Electrical Low Voltage Lead, CSU Formula SAE Electric (2025–2026) — led a 7+ engineer electrical/software team; secured New Eagle sponsorship.
- Safety Officer, RamBots CSU Senior Design (2025–2026) — owned safety protocols and documentation for a 13-person team.
- Ram Welcome Leader, CSU (Summer 2024) — mentored a group of 20 incoming students through campus transition.
- Member, National Society of Black Engineers (2023–2026) and ColorStack.

AWARDS & HONORS

Tesla Battery Sponsorship Recipient (2025) · Fry Family Electrical Engineering Scholarship · Carl Wilsen Scholarship in ECE · Engineering College Scholars · Dean's List (6 semesters)